

Turn Left / Turn Right Calibration R&D; Template

This template is used to calibrate 90° turning accuracy by adjusting motor PWM and TURN_BLIND_MS. Leave all values blank during planning and fill them in during testing.

Turn Right 90°

Function: `turnRight90()`

Objective: Determine the combination of motor PWM and TURN_BLIND_MS that produces the most accurate 90° rotation with minimum alignment error.

Experiment 1 - PWM Calibration

Keep TURN_BLIND_MS constant while changing the PWM values of all four motors.

Trial	UL PWM	LL PWM	UR PWM	LR PWM	TURN_BLIND_MS	Angular Error (°)	Observation
1							
2							
3							
4							
5							

Experiment 2 - TURN_BLIND_MS Calibration

Keep the PWM values constant while changing TURN_BLIND_MS.

Trial	UL PWM	LL PWM	UR PWM	LR PWM	TURN_BLIND_MS	Angular Error (°)	Observation
1							
2							
3							
4							
5							

Conclusion: Selected PWM = _____

Selected TURN_BLIND_MS = _____

Final Angular Error = _____

Turn Left 90°

Function: **turnLeft90()**

Objective: Determine the combination of motor PWM and TURN_BLIND_MS that produces the most accurate 90° rotation with minimum alignment error.

Experiment 1 - PWM Calibration

Keep TURN_BLIND_MS constant while changing the PWM values of all four motors.

Trial	UL PWM	LL PWM	UR PWM	LR PWM	TURN_BLIND_MS	Angular Error (°)	Observation
1							
2							
3							
4							
5							

Experiment 2 - TURN_BLIND_MS Calibration

Keep the PWM values constant while changing TURN_BLIND_MS.

Trial	UL PWM	LL PWM	UR PWM	LR PWM	TURN_BLIND_MS	Angular Error (°)	Observation
1							
2							
3							
4							
5							

Conclusion: Selected PWM = _____

Selected TURN_BLIND_MS = _____

Final Angular Error = _____